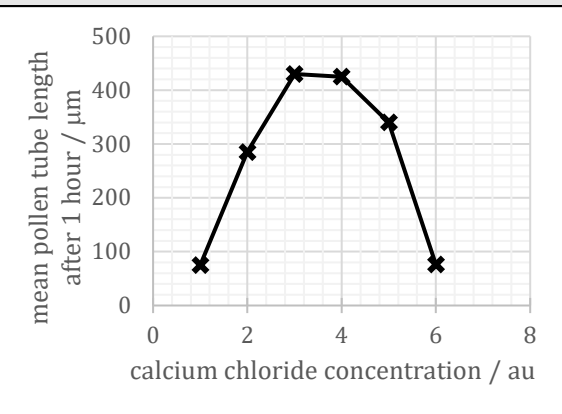


Question Number	Answer	Additional Guidance	Mark
2(a)	A description that includes the following points: • (digests a) pathway (through the style) / releases (digestive) enzymes (1) • for {male gametes / male nucleus generative nucleus(ii) / sperm} to the ovule (1)	IGNORE ovary, ACCEPT ovum, egg, female gamete	(2)

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	A graph showing the following features: • A axes correct (x - calcium chloride concentration / y - mean pollen tube length after 1 hour (1) • L axes correctly labelled and with units (1) • S linear scale on the x-axis (1) • P correct plotting on a linear scale on y (1)	 IGNORE line but if extrapolate to 0 on y loses mp P	(4)

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	A description that includes the following points: • use a range of 6 solutions of calcium chloride / from 1 – 6 au (1) • use of sucrose solution / boric acid (1) • controlled variable (1) • place pollen {on agar plate / in hanging drop} (1) • leave for 1 hour (1) • use a microscope and (calibrated) graticule to measure pollen tube length (1) • repeat readings at each calcium chloride concentration to allow calculation of mean (1)	ACCEPT CaCl_2, Ca ions, NOT Calcium e.g. temperature, pH, plant type, plant species NOT light intensity NOT use of Vernier calliper ACCEPT average	(6)

Question Number	Answer	Additional Guidance	Mark
2(b)(iii)	An answer that includes three of the following points: - pollen tubes do not grow at 1 au or 6 au (1) {tube growth is (nearly) constant between 3 and 4 au / {highest / optimum} growth {at 3 au / between 3 and 4 au / figure between 3 and 4 au quoted} (1) - the relationship is non-linear (1) - comment on {the lack of a control / no data for 0 au CaCl_2 / no SDS (1)	ACCEPT pollen tubes grow between 2 au and 5 au ACCEPT no growth at 1 au and very little a 6 au ACCEPT a description which implies non-linear ACCEPT error bars	(3)